Analysis on Seattle Weather Prediction Dataset

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The Seattle (city in USA) weather prediction dataset from Kaggle is used to analyze the weather conditions. It contains 1461 rows and 6 columns recorded from January 1st of 2012 to December 31st of 2015.

Among the 6 columns, two columns which are date and weather are categorical types and the remaining, which are precipitation, temp_max, temp_min and wind are numerical types.

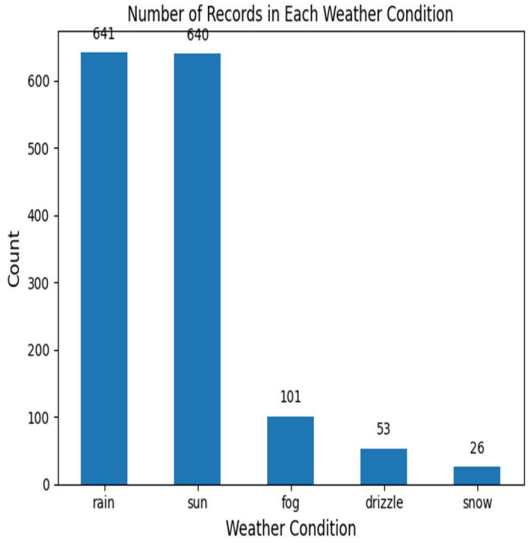


Fig 1: Bar Graph

The above Bar graph indicates that rain has the highest number of recordings, which is 641 followed very closely by sunny weather, with 640 observations and snow has the lowest recordings of 26.

So, from the graph concluded that the rainy and sunny conditions dominate in Seattle, whereas snowy conditions are comparatively rare.

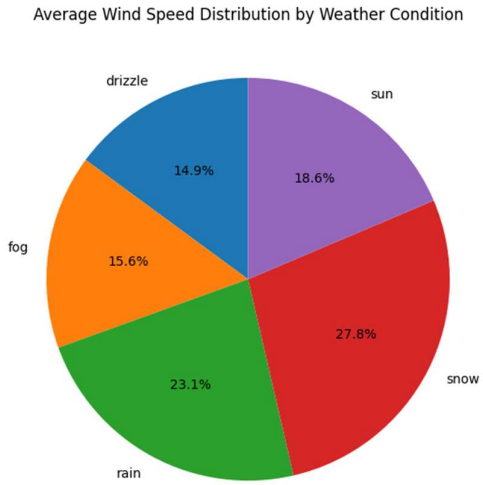


Fig 2: Pie Chart

The above Pie Chart shows the average wind speed distribution according to the weather conditions. The graph indicates that the snowy weather contributes the highest share of the average wind speeds that is 27.8% followed by the rainy weather at 23.1% and drizzle has the lowest share of 14.9%.

So, this indicates that snow and rain are often associated with stronger winds whereas drizzle has comparatively lower average wind speed.

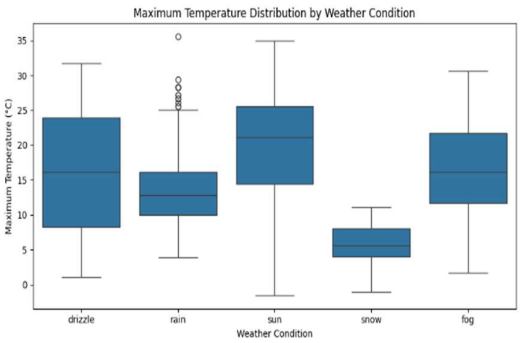


Fig 3: Box Plot

The above box plot represents the maximum temperature distribution by weather condition. It indicates that the highest temperature measured in sunny conditions as usual, that indicates warmer weather and snow has the lowest temperature indicating coldest weather. Overall sunny weather is associated with higher temperatures and

snowy conditions experience the lowest temperatures.

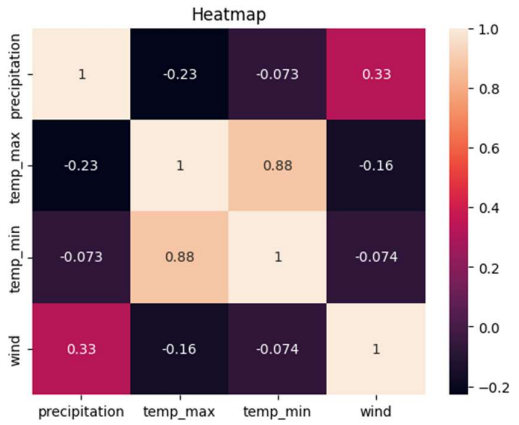


Fig 4: Heatmap

The above heatmap indicates the correlation between four features they are precipitation, maximum temperature, minimum temperature, and wind. Maximum and minimum temperatures have a strong positive correlation of 0.88, which indicates they increases together. And maximum temperature and precipitation has the negative correlation of -0.23 that means the precipitation related to the lower maximum temperature. Precipitation and wind have a weak positive correlation of 0.33.

Statistical Methods:

Mean ():

Mean gives the average of the data.

By using the formula:

Mean = Sum of observations/No. of observations

The average daily maximum temperature is approximately 16.44°C. The average precipitation recorded per day is approximately 3.03 units. The average daily maximum temperature is approximately 16.44°C. The mean indicates that the mean maximum temperature is higher than the mean minimum temperature and the mean difference between them is 8.20°C.

Median ():

The median represents the middle value of the data after arranging the data in ascending or descending order. The middle value of the daily maximum temperatures is 15.6°C. The middle value of the daily minimum temperatures is 8.3°C. The middle value of precipitation is zero that indicates that at least half of the recording days has no precipitation.

Standard Deviation:

The Standard Deviation is used to represents how much values in a dataset differ or spread out from their mean. The standard deviation value for the precipitation is 6.68, and for the maximum temperature is 7.35, and for the minimum temperature is 5.02, and for the wind is 1.44.

Skewness:

Skewness is a statistical measure that describes the shape of a data distribution. The skewness values indicate that the maximum temperature and precipitation has positive skewness that means those distributions have longer right tails and the minimum temperature has the neg skewness which is -0.24 which means the minimum temperature distributions has longer left tail.

Kurtosis:

Among all the features, Precipitation has the highest kurtosis value which is equal to 15.52, which indicates there is an observation of extreme rainfall and wind also shows the positive kurtosis of 0.8040 whereas the minimum and maximum temperatures have the negative kurtosis.